Changing climate

The blanket of gases that form Earth's atmosphere performs many valuable functions, from containing oxygen for respiration to shielding life from the Sun's harmful ultraviolet (UV) rays. It also traps heat, warming the planet's surface in a process called the greenhouse effect. In the past 200 years, a change to the balance of gases in the atmosphere has resulted in more heat being retained, causing global warming and climate change.

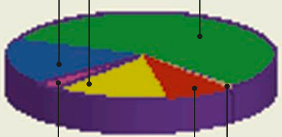
> **"Climate change is no longer some far-off problem... it is happening now."**
>
> **President Barack Obama, 2015**

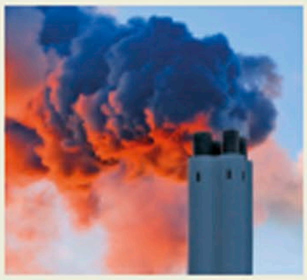
Shifting balance

Gases such as methane, carbon dioxide, and sulfur hexafluoride are known as greenhouse gases, and help create the greenhouse effect. Emissions caused by a booming human population and increasing impacts from industry, farming, and environmental damage have led to a rise in the concentrations of these gases in the atmosphere.

Greenhouse gas emissions

Carbon dioxide is the most common greenhouse gas emitted, with some 39.6 billion tons (36 billion metric tons) sent into the atmosphere each year. This gas is released from industry, from burning fossil fuels (natural gas, oil, and coal) in vehicle engines, and as part of electricity generation.





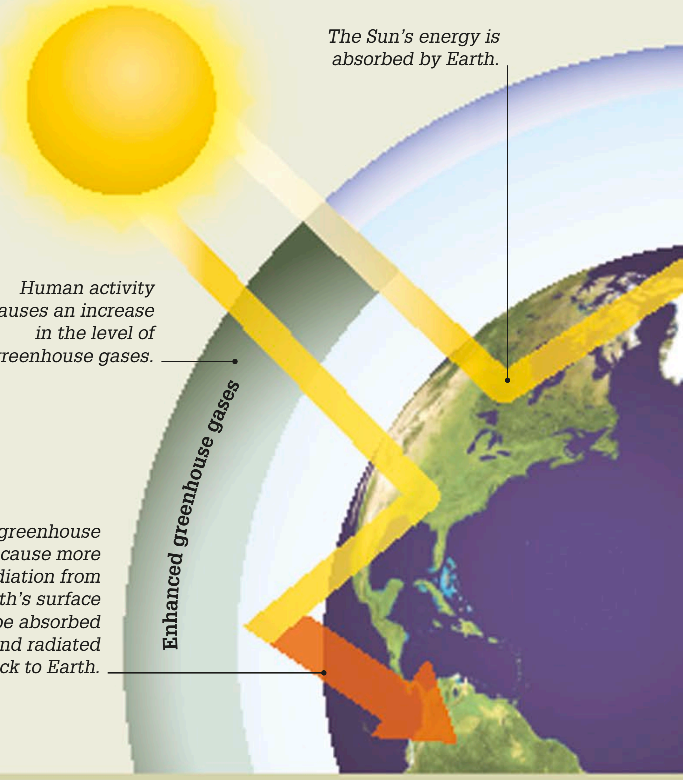
Many causes

The burning of fossil fuels in power stations (left) causes substantial carbon dioxide emissions. Other causes of emissions include cattle, which produce methane, and deforestation, which involves removing large numbers of trees that would normally absorb carbon dioxide.

Enhanced greenhouse gases

The enhanced greenhouse effect is caused by an increasing build-up of greenhouse gases in the atmosphere, which trap more heat than in the past, causing temperature rises on Earth. Temperatures will continue to increase if the level of greenhouse gases continues to grow.

Greenhouse gases in the atmosphere, showing the effect of increased gases (below, left)



Key events

1859

Irish physicist John Tyndall discovered how some gases block infrared radiation and suggested that changes in the concentration of atmospheric gases could affect the climate.

1958

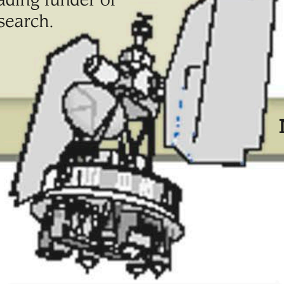
American scientist Charles David Keeling began a long-term study of carbon dioxide in the atmosphere. His Keeling Curve graph shows carbon dioxide rising from 310 parts per million (ppm) in 1958 to over 400 in 2015.

1970

The National Oceanic and Atmospheric Administration (NOAA) was founded in the US. It would become the world's leading funder of climate research.

1978

NASA launched the Scanning Multichannel Microwave Radiometer (SMMR) on the Nimbus satellite, to monitor sea ice in the Arctic and Antarctic.



Nimbus 7 satellite